5

Interpolation

Chapter Learning Outcomes

CLO

5

- Know and understand the meaning of interpolation.
- List different interpolation methods.
- Apply the Interpolation methods (Lagrange interpolation formula and Newton's divided-difference Method) for solving the problems numerically.

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Interpolation - Introduction

EXAMPLE

Based on the following data, find f(x) and f(2.5).

x	f(x)
1	2
2	5
3	10
4	17
5	26

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Interpolation - Introduction

EXAMPLE

Based on the following data, find f(x) and f(0.15).

x	f(x)
0	0.0000
0.1	0.0998
0.2	0.1987
0.3	0.2955
0.4	0.3894

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Interpolation - Introduction

Definition 1

Interpolation: is the process of fitting a function through the given data.

Definition 2

If $f(x)$ is a continuous function on $[a, b]$ and $\varepsilon < 0$ then there exits a polynomial $p(x)$ such that $|f(x) - p(x)| < \varepsilon$ for all $x \in [a, b]$.

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Interpolation - Introduction

x	f(x)
.	.
.	.
.	.

f(x) is unknown

Polynomial

$$p(x) = a_n x^n + a_{n-1} x^{n-1} + \ldots + a_0$$

$$p(x) \cong f(x)$$

$$|f(x) - p(x)| < \varepsilon$$

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Interpolation Methods

1- Lagrange Method

2- Newton's Divided-Difference Method

3- Neville's Method

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Lagrange Method

Lagrange Method

Given the values of the function f(x) at *n+1* distinct points, then Lagrange interpolating polynomial p(x) is given by:

x	f(x)
x_0	f(x_0)
x_1	f(x_1)
x_2	f(x_2)
…	…
x_n	f(x_n)

$$p(x) = L_0\, f(x_0) + L_1\, f(x_1) + L_2\, f(x_2) + L_3\, f(x_3) + \ldots + L_n\, f(x_n)$$

$$L_0 = \frac{(x - x_1)(x - x_2)(x - x_3)\ldots(x - x_n)}{(x_0 - x_1)(x_0 - x_2)(x_0 - x_3)\ldots(x_0 - x_n)}$$

$$L_1 = \frac{(x - x_0)(x - x_2)(x - x_3)\ldots(x - x_n)}{(x_1 - x_0)(x_1 - x_2)(x_1 - x_3)\ldots(x_1 - x_n)}$$

$$L_2 = \frac{(x - x_0)(x - x_1)(x - x_3)\ldots(x - x_n)}{(x_2 - x_0)(x_2 - x_1)(x_2 - x_3)\ldots(x_2 - x_n)}$$

$$L_3 = \frac{(x - x_0)(x - x_1)(x - x_2)\ldots(x - x_n)}{(x_3 - x_0)(x_3 - x_1)(x_3 - x_2)\ldots(x_3 - x_n)}$$

Lagrange Method

General Formula:

X_i	$y_i = f(x_i)$
x_0	$y_0 = f(x_0)$
x_1	$y_1 = f(x_1)$
x_2	$y_2 = f(x_2)$
$\vdots$	$\vdots$
x_n	$y_n = f(x_n)$

$$p(x) = L_0 f(x_0) + L_1 f(x_1) + \cdots + L_n f(x_n)$$

$$p(x) = \sum_{k=0}^{n} L_k f(x_k)$$

$$L_k = \frac{(x - x_0)(x - x_1)\ldots(x - x_{k-1})(x - x_{k+1})\ldots(x - x_n)}{(x_k - x_0)(x_k - x_1)\ldots(x_k - x_{k-1})(x_k - x_{i+1})\ldots(x_k - x_n)}$$

$$L_k = \prod_{i=0, i\neq k}^{n} \frac{(x - x_i)}{(x_k - x_i)}$$

First Degree Lagrange Polynomial

Number of points (2)

x	f(x)
x_0	f(x_0)
x_1	f(x_1)

$$p(x) = L_0\, f(x_0) + L_1\, f(x_1)$$

$$L_0 = \frac{(x - x_1)}{(x_0 - x_1)}$$

$$L_1 = \frac{(x - x_0)}{(x_1 - x_0)}$$

Second Degree Lagrange Polynomial

Number of points (3)

x	f(x)
x_0	f(x_0)
x_1	f(x_1)
x_2	f(x_2)

$$p(x) = L_0\, f(x_0) + L_1\, f(x_1) + L_2\, f(x_2)$$

$$L_0 = \frac{(x - x_1)(x - x_2)}{(x_0 - x_1)(x_0 - x_2)}$$

$$L_1 = \frac{(x - x_0)(x - x_2)}{(x_1 - x_0)(x_1 - x_2)}$$

$$L_2 = \frac{(x - x_0)(x - x_1)}{(x_2 - x_0)(x_2 - x_1)}$$

Third Degree Lagrange Polynomial

$Number\ of\ points(4)$

x	f(x)
x_0	f(x_0)
x_1	f(x_1)
x_2	f(x_2)
x_3	f(x_3)

$$p(x) = L_0\, f(x_0) + L_1\, f(x_1) + L_2\, f(x_2) + L_3\, f(x_3)$$

$$L_0 = \frac{(x - x_1)(x - x_2)(x - x_3)}{(x_0 - x_1)(x_0 - x_2)(x_0 - x_3)}$$

$$L_1 = \frac{(x - x_0)(x - x_2)(x - x_3)}{(x_1 - x_0)(x_1 - x_2)(x_1 - x_3)}$$

$$L_2 = \frac{(x - x_0)(x - x_1)(x - x_3)}{(x_2 - x_0)(x_2 - x_1)(x_2 - x_3)}$$

$$L_3 = \frac{(x - x_0)(x - x_1)(x - x_2)}{(x_3 - x_0)(x_3 - x_1)(x_3 - x_2)}$$

Lagrange Method

EXAMPLE

Based on the following data, Find the first degree (Linear) Lagrange interpolating polynomial.

x	f(x)
1	4
2	5

$$p(x) = L_0\, f(x_0) + L_1\, f(x_1)$$

$$L_0 = \frac{(x - x_1)}{(x_0 - x_1)}$$

$$L_1 = \frac{(x - x_0)}{(x_1 - x_0)}$$

$$p(x) = x + 3$$

Lagrange Method

EXAMPLE

Interpolate the following data using Lagrange interpolation and approximate f(0).

x	f(x)
-1	-3
1	-1
2	3

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Lagrange Method

x	f(x)
-1	-3
1	-1
2	3

$$p(x) = L_0 f(x_0) + L_1 f(x_1) + L_2 f(x_2)$$

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Lagrange Method

x	f(x)
-1	-3
1	-1
2	3

$$p(x) = L_0\, f(x_0) + L_1\, f(x_1) + L_2\, f(x_2)$$

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Lagrange Method

x	f(x)
-1	-3
1	-1
2	3

$$p(x) = L_0\, f(x_0) + L_1\, f(x_1) + L_2\, f(x_2)$$

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Lagrange Method

$$p(x) = L_0\, f(x_0) + L_1\, f(x_1) + L_2\, f(x_2)$$

x	f(x)
-1	-3
1	-1
2	3

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Lagrange Method

$$p(x) = L_0\, f(x_0) + L_1\, f(x_1) + L_2\, f(x_2)$$

x	f(x)
-1	-3
1	-1
2	3

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Lagrange Method

EXAMPLE

Find the second degree Lagrange interpolating polynomial for the function $f(x) = 2^x$ on [0, 2].

x	f(x)
0	1
1	2
2	4

Lagrange Method

x	f(x)
0	1
1	2
2	4

$$p(x) = L_0 f(x_0) + L_1 f(x_1) + L_2 f(x_2)$$

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Lagrange Method

x	f(x)
0	1
1	2
2	4

$$p(x) = L_0\, f(x_0) + L_1\, f(x_1) + L_2\, f(x_2)$$

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Lagrange Method

x	f(x)
0	1
1	2
2	4

$$p(x) = L_0\, f(x_0) + L_1\, f(x_1) + L_2\, f(x_2)$$

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Lagrange Method

$$p(x) = L_0\, f(x_0) + L_1\, f(x_1) + L_2\, f(x_2)$$

x	f(x)
0	1
1	2
2	4

Interpolation Polynomial for → $f(x) = 2^x$

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Lagrange Method

EXAMPLE

Based on the following data, approximate the value of f(1.25) using the Lagrange interpolating polynomial.

x	f(x)
0	1
1	6
3	34
7	162

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Lagrange Method

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Interpolation Methods

1- Lagrange Method

2- Newton's Divided-Difference Method

3- Neville's Method

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Newton's Divided-Difference Method

Newton's Divided-Difference Method

Given the values of the function f(x) at n+1 data points, then then the Newton's divided-difference polynomial p(x) is given by:

$$
\begin{aligned}
\boldsymbol{p(x)} = {} & F[x_0] \\
& + F[x_0, x_1]\,(\boldsymbol{x} - \boldsymbol{x_0}) \\
& + F[x_0, x_1, x_2]\,(\boldsymbol{x} - \boldsymbol{x_0})(\boldsymbol{x} - \boldsymbol{x_1}) \\
& + F[x_0, x_1, x_2, x_3]\,(\boldsymbol{x} - \boldsymbol{x_0})(\boldsymbol{x} - \boldsymbol{x_1})(\boldsymbol{x} - \boldsymbol{x_2}) \\
& + . \\
& + .
\end{aligned}
$$

Newton's Divided-Difference Method

Zeroth Divided-Difference $F[x_i]$.

$$F[x_i]=f(x_i)$$

First Divided-Difference

$$F[x_i, x_{i+1}]=\frac{F[x_{i+1}]-F[x_i]}{x_{i+1}-x_i}$$

Newton's Divided-Difference Method

Second Divided-Difference

$$F[x_i, x_{i+1}, x_{i+2}] = \frac{F[x_{i+1}, x_{i+2}] - F[x_i, x_{i+1}]}{x_{i+2} - x_i}$$

Kth Divided-Difference

$$F[x_i, x_{i+1}, \ldots, x_{i+k}] = \frac{F[x_{i+1}, \ldots, x_{i+k}] - F[x_i, \ldots, x_{i+k-1}]}{x_{i+k} - x_i}$$

Divided-Difference Table

x	0th D.D	1st D.D	2nd D.D	3rd D.D
	F[ ]	F[,]	F[, ,]	F[, , ,]
x_0	$F[x_0]$	$F[x_0,x_1]$	$F[x_0,x_1,x_2]$	$F[x_0,x_1,x_2,x_3]$
x_1	$F[x_1]$	$F[x_1,x_2]$	$F[x_1,x_2,x_3]$	
x_2	$F[x_2]$	$F[x_2,x_3]$		
x_3	$F[x_3]$			

Newton's Divided-Difference Method

Example: Find the second degree Newton's interpolating polynomial for the following Data.

x	f(x)
0	-5
1	-3
-1	-15

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Divided-Difference Table

x	0th D.D	1st D.D	2nd D.D
	F[ ]	F[,]	F[, ,]
x_0	$F[x_0]=$	$F[x_0,x_1]=$	$F[x_0,x_1,x_2]=$
x_1	$F[x_1]=$	$F[x_1,x_2]=$	
x_2	$F[x_2]=$		

x	f(x)
0	-5
1	-3
-1	-15

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Divided-Difference Table

x	0th D.D	1st D.D	2nd D.D
	F[ ]	F[,]	F[, ,]
x_0	$F[x_0]=$	$F[x_0,x_1] =$	$F[x_0,x_1,x_2] =$
x_1	$F[x_1]=$	$F[x_1,x_2] =$	
x_2	$F[x_2]=$		

x	f(x)
0	-5
1	-3
-1	-15

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Divided-Difference Table

x	0th D.D	1st D.D	2nd D.D
	F[ ]	F[,]	F[, ,]
x_0	$F[x_0]=$	$F[x_0,x_1] =$	$F[x_0,x_1,x_2]=$
x_1	$F[x_1]=$	$F[x_1,x_2] =$	
x_2	$F[x_2]=$		

x	f(x)
0	-5
1	-3
-1	-15

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Divided-Difference Table

x	0th D.D	1st D.D	2nd D.D
	F[ ]	F[,]	F[, ,]
x_0	$F[x_0]=$	$F[x_0,x_1] =$	$F[x_0,x_1,x_2] =$
x_1	$F[x_1]=$	$F[x_1,x_2] =$	
x_2	$F[x_2]=$		

x	f(x)
0	-5
1	-3
-1	-15

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Divided-Difference Table

x	0th D.D	1st D.D	2nd D.D
	F[ ]	F[,]	F[, ,]
x_0	$F[x_0]=$	$F[x_0,x_1] =$	$F[x_0,x_1,x_2] =$
x_1	$F[x_1]=$	$F[x_1,x_2] =$	
x_2	$F[x_2]=$		

x	f(x)
0	-5
1	-3
-1	-15

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Newton's Divided-Difference Method

$p(x) =$

x	f(x)
0	-5
1	-3
-1	-15

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Newton's Divided-Difference Method

Example: Based on the following data, approximate f(1.5) using the Newton's divided difference interpolation method.

x	f(x)
-1	-21
1	15
2	12
3	3

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Divided-Difference Table

x	0th D.D	1st D.D	2nd D.D	3rd D.D
	F[ ]	F[,]	F[, ,]	F[, , ,]
x_0	$F[x_0]=$	$F[x_0,x_1]=$	$F[x_0,x_1,x_2]=$	$F[x_0,x_1,x_2,x_3]=$
x_1	$F[x_1]=$	$F[x_1,x_2]=$	$F[x_1,x_2,x_3]=$	
x_2	$F[x_2]=$	$F[x_2,x_3]=$		
x_3	$F[x_3]=$			

x	f(x)
-1	-21
1	15
2	12
3	3

Divided-Difference Table

x	0th D.D	1st D.D	2nd D.D	3rd D.D
	F[ ]	F[,]	F[, ,]	F[, , ,]
x_0	$F[x_0]=$	$F[x_0,x_1]=$	$F[x_0,x_1,x_2]=$	$F[x_0,x_1,x_2,x_3]=$
x_1	$F[x_1]=$	$F[x_1,x_2]=$	$F[x_1,x_2,x_3]=$	
x_2	$F[x_2]=$	$F[x_2,x_3]=$		
x_3	$F[x_3]=$			

x	f(x)
-1	-21
1	15
2	12
3	3

$F[x_0,x_1]=$

معيدي

Divided-Difference Table

x	0th D.D	1st D.D	2nd D.D	3rd D.D
	F[ ]	F[,]	F[, ,]	F[, , ,]
x_0	$F[x_0]=$	$F[x_0,x_1]=$	$F[x_0,x_1,x_2]=$	$F[x_0,x_1,x_2,x_3]=$
x_1	$F[x_1]=$	$F[x_1,x_2]=$	$F[x_1,x_2,x_3]=$	
x_2	$F[x_2]=$	$F[x_2,x_3]=$		
x_3	$F[x_3]=$			

x	f(x)
-1	-21
1	15
2	12
3	3

$F[x_1,x_2]=$

Divided-Difference Table

x	0th D.D	1st D.D	2nd D.D	3rd D.D
	F[ ]	F[,]	F[, ,]	F[, , ,]
x_0	$F[x_0]=$	$F[x_0,x_1]=$	$F[x_0,x_1,x_2]=$	$F[x_0,x_1,x_2,x_3]=$
x_1	$F[x_1]=$	$F[x_1,x_2]=$	$F[x_1,x_2,x_3]=$	
x_2	$F[x_2]=$	$F[x_2,x_3]=$		
x_3	$F[x_3]=$			

x	f(x)
-1	-21
1	15
2	12
3	3

$F[x_2,x_3]=$

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Divided-Difference Table

x	0th D.D	1st D.D	2nd D.D	3rd D.D
	F[ ]	F[,]	F[, ,]	F[, , ,]
x_0	$F[x_0]=$	$F[x_0,x_1]=$	$F[x_0,x_1,x_2]=$	$F[x_0,x_1,x_2,x_3]=$
x_1	$F[x_1]=$	$F[x_1,x_2]=$	$F[x_1,x_2,x_3]=$	
x_2	$F[x_2]=$	$F[x_2,x_3]=$		
x_3	$F[x_3]=$			

x	f(x)
-1	-21
1	15
2	12
3	3

$F[x_0,x_1,x_2]=$

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Divided-Difference Table

x	0th D.D	1st D.D	2nd D.D	3rd D.D
	F[ ]	F[,]	F[, ,]	F[, , ,]
x_0	$F[x_0]=$	$F[x_0,x_1]=$	$F[x_0,x_1,x_2]=$	$F[x_0,x_1,x_2,x_3]=$
x_1	$F[x_1]=$	$F[x_1,x_2]=$	$F[x_1,x_2,x_3]=$	
x_2	$F[x_2]=$	$F[x_2,x_3]=$		
x_3	$F[x_3]=$			

x	f(x)
-1	-21
1	15
2	12
3	3

$F[x_1,x_2,x_3]=$

Divided-Difference Table

x	0th D.D	1st D.D	2nd D.D	3rd D.D
	F[ ]	F[,]	F[, ,]	F[, , ,]
x_0	$F[x_0]=$	$F[x_0,x_1]=$	$F[x_0,x_1,x_2]=$	$F[x_0,x_1,x_2,x_3]=$
x_1	$F[x_1]=$	$F[x_1,x_2]=$	$F[x_1,x_2,x_3]=$	
x_2	$F[x_2]=$	$F[x_2,x_3]=$		
x_3	$F[x_3]=$			

x	f(x)
-1	-21
1	15
2	12
3	3

$F[x_0,x_1,x_2,x_3]=$

Divided-Difference Table

x	0th D.D	1st D.D	2nd D.D	3rd D.D
	F[ ]	F[,]	F[, ,]	F[, , ,]
x_0	$F[x_0]=$	$F[x_0,x_1]=$	$F[x_0,x_1,x_2]=$	$F[x_0,x_1,x_2,x_3]=$
x_1	$F[x_1]=$	$F[x_1,x_2]=$	$F[x_1,x_2,x_3]=$	
x_2	$F[x_2]=$	$F[x_2,x_3]=$		
x_3	$F[x_3]=$			

x	f(x)
-1	-21
1	15
2	12
3	3

Newton's Divided-Difference Method

x	f(x)
-1	-21
1	15
2	12
3	3

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Check - Chapter Learning Outcomes

CLO

5

- Know and understand the meaning of interpolation.
- List different interpolation methods.
- Apply the Interpolation methods (Lagrange interpolation formula and Newton's divided-difference Method) for solving the problems numerically.

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End of Chapter

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